Unit no. ° 2

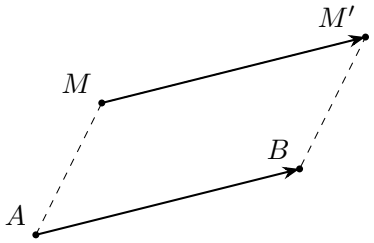
VECTORS IN THE PLANE

14-30 September 2026

I Translation and vectors

Definition

The translation that maps A to B assigns to every point M the unique point M' such that $ABM'M$ is a parallelogram, possibly degenerate. It is called the translation by vector $\overrightarrow{AB}$. A non-zero vector has a direction, a sense and a norm. $\|AB\| = AB$. The vector $\overrightarrow{AA}$ is the zero vector, denoted 0; it has no direction or sense.



Equal and opposite vectors

$\overrightarrow{AB} = \overrightarrow{CD}$ when the translation by vector $\overrightarrow{AB}$ maps C to D. Two non-zero vectors are equal if and only if they have the same direction, same sense and same norm. The opposite vector of AB is BA : $BA = -AB$.

Example 2.1

Draw a segment $[AB]$ and plot a point C not on (AB). Construct D such that $\overrightarrow{AB} = \overrightarrow{CD}$. Name the parallelogram obtained. Construct E such that $\overrightarrow{CE} = -\overrightarrow{AB}$.

Solution:

D is constructed by marking off from C the direction, the sense and the length of $\overrightarrow{AB}$. The parallelogram is $ABDC$, in this order. E is the reflection of D with respect to C, since $\overrightarrow{CE} = -CD$.

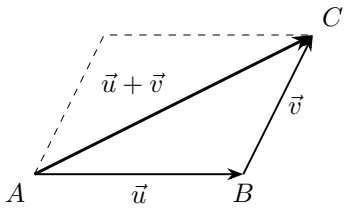
II Sum, difference and Chasles' relation

Sum of vectors

The sum $\vec{u}+\vec{v}$ corresponds to the sequence of translations by the vectors $\vec{u}$ and $\vec{v}$. For all points A, B, C :

$$\boxed{\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}}$$ (Chasles' relation).

We have $\vec{u}-\vec{v}=\vec{u}+(-\vec{v})$ and $\overrightarrow{AB}+\overrightarrow{BA}=\vec{0}$.



Parallelogram rule

$ABDC$ is a parallelogram if and only if $\overrightarrow{AD}=\overrightarrow{AB}+\overrightarrow{AC}$.

Example 2.2

Simplify: $\overrightarrow{AB}+\overrightarrow{BC}$; $\overrightarrow{AB}+\overrightarrow{CA}$; $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}$; $\overrightarrow{AB}-\overrightarrow{AC}$.

Solution:

$\overrightarrow{AC}$; $\overrightarrow{CA}+\overrightarrow{AB}=\overrightarrow{CB}$; $\overrightarrow{AD}$; $\overrightarrow{AB}+\overrightarrow{CA}=\overrightarrow{CB}$. Commutativity of addition allows us to reorder the vectors.

Example 2.3

In a parallelogram $ABCD$, justify that $\overrightarrow{AB}+\overrightarrow{AD}=\overrightarrow{AC}$.

Solution:

In $ABCD$, $\overrightarrow{AD}=\overrightarrow{BC}$. Hence $\overrightarrow{AB}+\overrightarrow{AD}=\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$.

III Product of a vector by a real number

Definition and calculation rules

For $k \in \mathbb{R}$, $k\vec{u} = \vec{0}$ if $k = 0$ or $\vec{u} = \vec{0}$. Otherwise, $k\vec{u}$ has the same direction as $\vec{u}$, the same sense if $k > 0$, the opposite sense if $k < 0$ and its norm is $|k|\|\vec{u}\|$.

$$(k+\ell)\vec{u} = k\vec{u} + \ell\vec{u}, \quad k(\vec{u}+\vec{v}) = k\vec{u} + k\vec{v}, \quad k(\ell\vec{u}) = (k\ell)\vec{u}.$$

Example 2.4

For a vector $\vec{u}$ of norm 4 cm, represent 2 $\vec{u}$, $-\vec{u}$ and $\frac{1}{2}\vec{u}$ from the same origin. Specify their norms.

Solution:

The norms are 8 cm, 4 cm, 2 cm. The vectors 2 $\vec{u}$ and $\frac{1}{2}\vec{u}$ have the sense of $\vec{u}$; $-\vec{u}$ has the opposite sense.

Characterisations of the midpoint

I is the midpoint of $[AB]$ if and only if $\overrightarrow{AI} = \overrightarrow{IB}$, or else $\overrightarrow{AI} = \frac{1}{2}\overrightarrow{AB}$, or $\overrightarrow{IA} + \overrightarrow{IB} = \vec{0}$.

Example 2.5

In a triangle ABC, I is the midpoint of $[AB]$ and J that of $[AC]$. Show that $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{BC}$.

Solution:

By Chasles, $\overrightarrow{IJ} = \overrightarrow{IA} + \overrightarrow{AJ} = -\frac{1}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AC} = \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2}\overrightarrow{BC}$. We recover the parallelism of the lines (IJ) and (BC) when the triangle is not degenerate.

Fluency · 5 to 10 minutes

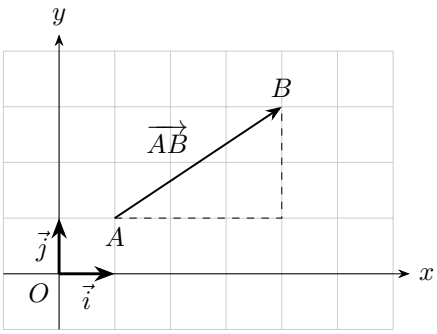
Simplify 3 $\vec{u} - 5\vec{u}$; calculate $(-3)^2$; solve 2 $x + 1 = 7$.

Answers: $-2\vec{u}$; 9 ; $x = 3$.

IV Coordinate systems, bases and coordinates

Definitions

Two non-collinear vectors $\vec{i}, \vec{j}$ form a basis of the plane. Every vector $\vec{u}$ can be written in a only as $\vec{u} = x\vec{i} + y\vec{j}$; its coordinates are ($x; y$) or $\begin{pmatrix} x \\ y \end{pmatrix}$. A coordinate system $(O; \vec{i}, \vec{j})$ associates with the point M the coordinates of $\overrightarrow{OM}$. It is **orthonormal** if the basis vectors are perpendicular and of norm 1.



So $\overrightarrow{AB} = 3\vec{i} + 2\vec{j}$: we move three units to the right and two units up.

Calculations with coordinates

If $A(x_A; y_A)$, $B(x_B; y_B)$, $\vec{u}(x; y)$ and $\vec{v}(x'; y')$:

$$\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}, \quad \vec{u} + \vec{v} = \begin{pmatrix} x + x' \\ y + y' \end{pmatrix}, \quad k\vec{u} = \begin{pmatrix} kx \\ ky \end{pmatrix}.$$

Two vectors are equal if and only if their respective coordinates are equal.

Example 2.6

For $\vec{u}(3; 2)$ and $\vec{v}(-1; 4)$, calculate $\vec{u} + \vec{v}$ and $2\vec{u} - \vec{v}$. Write $\vec{w}(5; -3)$ in the basis ($i, \vec{j}$).

Solution:

> $\vec{u} + \vec{v} = (2; 6)$; $2\vec{u} - \vec{v} = (7; 0)$; $\vec{w} = 5\vec{i} - 3\vec{j}$.

V Norm, distance and midpoint

Properties

In an **orthon. frame**, for $\vec{u}(x;y)$ and two points A, B :

$$\|\vec{u}\| = \sqrt{x^2+y^2}, \qquad AB = \sqrt{(x_B-x_A)^2+(y_B-y_A)^2}.$$

In any coordinate system, the midpoint of $[AB]$ is $I\left(\frac{x_A+x_B}{2}; \frac{y_A+y_B}{2}\right)$.

Write a proof.

Solution:

The distance formula follows from Pythagoras' theorem applied to the horizontal and vertical moves. Midpoint, $AI = \frac{1}{2}AB$ gives $x_I - x_A = \frac{x_B-x_A}{2}$ & $y_I - y_A = \frac{y_B-y_A}{2}$; solve for x_I, y_I.

Example 2.7

In an orthonormal coordinate system, $A(3;0)$, $B(5;3)$, $C(4;0)$ and $D(6;4)$.

1. Calculate $\overrightarrow{AB}$, $\overrightarrow{CD}$, then $3\overrightarrow{AB} - 5\overrightarrow{CD}$.
2. Determine U such that $\overrightarrow{BU} = 2\overrightarrow{AB} - \overrightarrow{CD}$.
3. Determine E such that $ABCE$ is a parallelogram, then its centre I.
4. Calculate AB and CD.

Solution:

1. $\overrightarrow{AB}$ = (2; 3) and $\overrightarrow{CD}$ = (2; 4), so $3\overrightarrow{AB} - 5\overrightarrow{CD} = (-4;-11)$.
2. $\overrightarrow{BU}$ = (2; 2), so $U = B + (2;2) = (7;5)$.
3. $E = A + C - B = (2;-3)$. The centre is the midpoint of $[AC]$: $I(7/2;0)$. The midpoint of $[BE]$ gives the same result.
4. $AB = \sqrt{13}$ and $CD = \sqrt{20} = 2\sqrt{5}$.

VI Collinearity and determinant

Definition

Two non-zero vectors are collinear if they have the same direction. The zero vector is collinear with every vector. If $\vec{v} \neq \vec{0}$, $\vec{u}$ and $\vec{v}$ are collinear if and only if there is a real number k such that $\vec{u} = k\vec{v}$.

Watch out. In the last characterisation, the condition $\vec{v} \neq \vec{0}$ is necessary: a non-zero vector cannot be a multiple of the zero vector.

Determinant and collinearity criterion

For $\vec{u}(x; y)$, $\vec{v}(x'; y')$, $\det(\vec{u}, \vec{v}) = xy' - yx'$.

$$\vec{u}, \vec{v} \text{ collinear} \iff \det(\vec{u}, \vec{v}) = 0.$$

Write a proof.

Solution:

If $\vec{u} = k\vec{v}$, then $xy' - yx' = kx'y' - ky'x' = 0$. The case where a vector is zero is immediate. Conversely, suppose $\vec{v} \neq \vec{0}$ and $xy' - yx' = 0$. If $x' \neq 0$, let $k = x/x'$. Equality gives $y = ky'$, therefore $\vec{u} = k\vec{v}$. If $x' = 0$, then $y' \neq 0$, hence $x = 0$; with $k = y/y'$, we still get $\vec{u} = k\vec{v}$.

Example 2.8

Study collinearity: (a) $\vec{u}(3; 2)$ and $\vec{v}(-6; -5)$; (b) $\vec{w}(4; 1)$ and $\vec{t}(3; 3/4)$. If $\overrightarrow{AB} = 2\overrightarrow{CD}$ and $\overrightarrow{BD} = -5\overrightarrow{CD}$, relate $\overrightarrow{AB}$ and $\overrightarrow{BD}$.

Solution:

(a) $\det(\vec{u}, \vec{v}) = -15 + 12 = -3 \neq 0$: not collinear. (b) $\vec{t} = \frac{3}{4}\vec{w}$: collinear; the determinant is $4 \times \frac{3}{4} - 3 = 0$. Finally $\overrightarrow{AB} = -\frac{2}{5}\overrightarrow{BD}$, including if the vectors are zero.

VII Collinearity and parallelism

Applications

For $A \neq B$ and $C \neq D$, the lines (AB) and (CD) are parallel (possibly coincident) if and only if $\det(\overrightarrow{AB}, \overrightarrow{CD}) = 0$. The points A, B, C are collinear if and only if $\det(\overrightarrow{AB}, \overrightarrow{AC}) = 0$.

Example 2.9

Show that (AB) and (CD) are parallel with $A(2; -4)$, $B(10; -2)$, $C(-16; -4)$ and $D(12; 3)$.

Solution:

$\overrightarrow{AB}$ = (8; 2) and $\overrightarrow{CD}$ = (28; 7). Their determinant is $8 \times 7 - 2 \times 28 = 0$. The points of each pair are distinct: the two lines exist and are parallel.

Example 2.10

Show that $E(0; 1/2)$, F(2; 1) and G(6; 2) are collinear.

Solution:

$\overrightarrow{EF} = (2; 1/2)$ and $\overrightarrow{EG} = (6; 3/2) = 3\overrightarrow{EF}$. The determinant equals $2 \times 3/2 - (1/2) \times 6 = 0$; therefore the points are collinear.

Example 2.11

With $A(-1; 1)$, $B(3; 2)$, $C(-2; -3)$, $D(6; -1)$, E(5; 0), show that $(AB) \parallel (CD)$ and that E, B, D are collinear.

Solution:

$\overrightarrow{AB}$ = (4; 1) and $\overrightarrow{CD} = (8; 2) = 2\overrightarrow{AB}$. Then $\overrightarrow{EB} = (-2; 2)$ and $\overrightarrow{ED} = (1; -1) = -\frac{1}{2}\overrightarrow{EB}$. Both results follow.